

# YADNIK BENDALE

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City of Birth: Mumbai | Country of Birth: India

## Education

**University of Pennsylvania**, School of Engineering and Applied Science, Philadelphia Aug 2023 – May 2025  
*Master of Science in Engineering: Electrical Engineering Specialization in Circuits and Computer Engineering* (GPA: 3.77/4.0)

**Dwarkadas J. Sanghvi College of Engineering**, Mumbai, India Aug 2018 – Jun 2022  
*Bachelor of Engineering in Electronics and Telecommunication Engineering* (GPA: 8.71/10)

## Technical Skills

- **Programming and Scripting Languages:** C, C++, Python, ARM Assembly
- **Protocols:** UART, SPI, I2C, RS485, CAN, PWM, PCIe, Bluetooth Low Energy, Zigbee, MQTT, LoRa
- **Hardware:** STM32H753ZI Nucleo, TM4C123GXL Tiva C, Teensy 3.6, ATSAM21-XP, CC1352P Launchpad, nRF7002-DK, Raspberry Pi, BeagleBone Black, ESP32 C6, Xbee Pro S2C, Digital Multimeter, Oscilloscope, Logic Analyzer, JTAG
- **Software Tools:** MATLAB, Git, Jira, IAR, STM32CubeIDE, Keil µVision, Code Composer Studio, Altium Designer

## Professional Experience

**Torch Systems**, *Firmware Engineer*, San Francisco, California Feb 2026 – Present

- Engineered a 2-stage FreeRTOS-based on-device AI detection pipeline on STM32U5, fusing dual-exposure HDR frames across 3 camera channels through a motion score grid model and MobileNetV3 human classifier, achieving sub-200ms end-to-end inference latency.
- Architected a 17-sensor FreeRTOS driver framework spanning environmental, power, motion, and imaging peripherals over I2C, delivering encrypted JSON telemetry via TLS to cloud over Sierra HL7802 LTE-M across intermittent cellular conditions.
- Designed and deployed OTA firmware update system using AWS IoT Jobs with cryptographic image verification, automatic rollback, and watchdog crash recovery, achieving 98% successful update delivery across concurrent firmware versions.

**ePlant Inc**, *Firmware Engineer*, San Francisco, California May 2025 – Feb 2026

- Designed a C-based firmware optimization suite reducing device boot time by 40%, implementing Adaptive Data Rate and enforcing protocol limits, improving battery life by 30%.
- Engineered FUOTA delivery over LoRaWAN with multi-slot bootloader featuring automatic failure recovery and reboot cause tracking, reducing field failure interventions by 80%.

**Span.IO**, *Device Software Engineer Intern*, San Francisco, USA Jun 2024 – Aug 2024

- Automated HIL testing for SPAN Panel modules by integrating Nexio with Buildkite CI/CD pipelines, enabling automated firmware flashing over CAN and nightly build tests, reducing manual testing time by 60%.
- Customized secure bootloader and OTA provisioning for STM32-based SPAN modules, enabling ARM TrustZone secure boot, dual-slot A/B firmware management, and cryptographic firmware signing.

**Devlata Technologies Pvt Ltd**, *Embedded System Developer*, Mumbai, India Jan 2023 – Jun 2023

- Developed STM32-based firmware in C for safety-critical overhead crane encoder/decoder units, optimizing timer-driven interrupts to achieve sub-2ms transmission latency.
- Enhanced RF communication reliability by 45% using NRF905 transceiver at 433 MHz, improving range and robustness.

**Indian Institute of Technology Bombay**, *Embedded Systems Project Associate*, Mumbai, India Jul 2022 – Dec 2022

- Developed telematics device enabling user-based insurance models, reducing insurance premiums by up to 30%.
- Achieved 20% IMU noise reduction using Allan Variance analysis and Savitzky-Golay filtering in Python, improving sensor accuracy for real-time vehicle dynamics monitoring.

**Google Summer of Code, BeagleBoard.org**, *Student Developer*, Remote May 2021 – Aug 2021

- Ported BeagleConnect Freedom to Zephyr RTOS by developing CMake, Kconfig, and Device Tree configurations, achieving 15% improvement in boot time.
- Developed MicroPython drivers for 8 MIKROE click boards using I2C, SPI, GPIO, and UART, enabling dual runtime with Zephyr RTOS and MicroPython.

## Projects

**SafeScan – RFID-Based Retained Foreign Object Tracking System** ([GitHub Link](#)) Sep 2024 – Dec 2024

- Developed RFID-based surgical instrument tracking system on nRF7002DK using Zephyr RTOS and AWS IoT Core for real-time inventory monitoring to prevent Retained Foreign Objects during surgeries.
- Implemented multi-layer security with ARM TrustZone for hardware isolation, I2C device whitelisting, and end-to-end encryption using MQTT over TLS, ensuring 95% tracking accuracy.

**Wireless Logic Analyzer** ([GitHub Link](#)) Feb 2024 – May 2024

- Designed a 4-layer PCB with SAMW25 and developed FreeRTOS-based firmware in C to support multi-protocol sampling.
- Achieved sub-millisecond timing accuracy using 800 kHz ADC and RTC, with real-time data streaming to Node-RED dashboard over MQTT, improving debugging efficiency by 35%.